CS24303 • MODULE 2

Module 2: Data Preprocessing & Dimensionality Reduction

Topics 15 to 20 • Missing Imputation, Binning, Chi-Square Independence, Normalization & PCA

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Topic 15 & 16: Data Cleaning, Missing Value Imputation & Binning Smoothing

Real-world operational data is dirty: incomplete (lacking attribute values), noisy (containing errors or outliers), and inconsistent (containing discrepancies in codes or names). **Data Preprocessing** consumes up to 80% of the total time in data mining projects.

Missing Data Strategy	Algorithmic Mechanism	Trade-offs & Best Use Cases
Ignore the Tuple	Discard any row containing missing values.	Drastically reduces sample size; only acceptable when missingness is $< 1\%$ and missing completely at random (MCAR).
Global Constant Imputation	Replace missing values with a placeholder string (e.g., `"Unknown"` or `-\infty`).	Simple, but distorts statistical distributions and introduces spurious constants into tree splits.
Attribute Mean / Median	Replace missing values with the arithmetic mean $\bar{x}$ (for symmetric data) or median (for skewed data).	Preserves sample size; artificially deflates attribute variance and alters inter-feature covariances.
Class-Conditional Mean / Median	Replace missing value with the mean/median of the subset belonging to the *same target class label*.	Significantly more accurate; preserves class separability in supervised learning tasks.
Model-Based Imputation (k-NN / MICE / EM)	Treat missing attribute as a target variable; predict its value using regression, k-NN neighbors, or Multiple Imputation by Chained Equations.	State-of-the-art accuracy; computationally expensive for massive streaming datasets.

🏛 Step-by-Step Solved Problem: Data Smoothing by Binning

Consider the sorted price dataset ($N = 12$):

$$\mathbf{X} = \{\mathbf{4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34}\}$$

Partition into 3 equal-frequency bins (Bin Depth = 4 items per bin):

- **Bin 1:** $\{4, 8, 9, 15\}$
- **Bin 2:** $\{21, 21, 24, 25\}$
- **Bin 3:** $\{26, 28, 29, 34\}$

1. Smoothing by Bin Means:

- Bin 1 Mean = $\frac{4+8+9+15}{4} = \frac{36}{4} = \mathbf{9} \implies \{9, 9, 9, 9\}$
- Bin 2 Mean = $\frac{21+21+24+25}{4} = \frac{91}{4} = \mathbf{22.75} \implies \{22.75, 22.75, 22.75, 22.75\}$
- Bin 3 Mean = $\frac{26+28+29+34}{4} = \frac{117}{4} = \mathbf{29.25} \implies \{29.25, 29.25, 29.25, 29.25\}$

2. Smoothing by Bin Boundaries:

Replace each value with the closer boundary (minimum or maximum of the bin):

- Bin 1 ($[\min = 4, \max = 15]$): $4 \to 4,\ 8 \to 4\ (|8-4| = 4 < |8-15| = 7),\ 9 \to 15\ (|9-15| = 6 < |9-4| = 5 \text{ wait}, |9-4| = 5 \implies 4),\ 15 \to 15 \implies \{\mathbf{4, 4, 4, 15}\}$
- Bin 2 ($[\min = 21, \max = 25]$): $21 \to 21,\ 21 \to 21,\ 24 \to 25,\ 25 \to 25 \implies \{\mathbf{21, 21, 25, 25}\}$
- Bin 3 ($[\min = 26, \max = 34]$): $26 \to 26,\ 28 \to 26,\ 29 \to 26,\ 34 \to 34 \implies \{\mathbf{26, 26, 26, 34}\}$

Topic 17 & 18: Data Integration (χ^2, Pearson r) & Normalization

Correlation & Normalization Mathematical Formulations:

$$\chi^2 \text{ Chi-Square Test of Independence: } \chi^2 = \sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \qquad E_{ij} = \frac{\text{row\_total}_i \times \text{col\_total}_j}{N}$$

$$\text{Pearson Product-Moment Correlation: } r_{A,B} = \frac{\sum_{i=1}^{N}(a_i - \bar{A})(b_i - \bar{B})}{(N-1)s_A s_B} = \frac{\text{Cov}(A,B)}{s_A s_B}$$

$$\text{Min-Max Normalization to } [\text{new\_min}, \text{new\_max}] : v' = \frac{v - \min_A}{\max_A - \min_A}(\text{new\_max} - \text{new\_min}) + \text{new\_min}$$

$$\text{Z-Score Standardization (Zero-Mean, Unit-Variance): } v' = \frac{v - \bar{A}}{\sigma_A} \quad (\text{Robust to outliers: } v' = \frac{v - \text{Median}}{\text{Mean Absolute Deviation}})$$

$$\text{Decimal Scaling Normalization: } v' = \frac{v}{10^j} \quad (\text{where } j \text{ is the smallest integer s.t. } \max(|v'|) < 1)$$

🏛 Step-by-Step Solved Problem: χ^2 Test of Independence on Contingency Table

Test whether gender is independent of preferred news medium at $\alpha = 0.05$ significance level:

Gender \ Medium	Digital News	Print Newspaper	Total
Male	$O_{11} = 250$	$O_{12} = 50$	300
Female	$O_{21} = 200$	$O_{22} = 100$	300
Total	450	150	$\mathbf{N = 600}$

1. Calculate Expected Frequencies ($E_{ij} = \frac{\text{RowTotal} \times \text{ColTotal}}{N}$):

- $E_{11} = \frac{300 \times 450}{600} = \mathbf{225}$ $\quad E_{12} = \frac{300 \times 150}{600} = \mathbf{75}$
- $E_{21} = \frac{300 \times 450}{600} = \mathbf{225}$ $\quad E_{22} = \frac{300 \times 150}{600} = \mathbf{75}$

2. Compute χ^2 Statistic:

$$\chi^2 = \frac{(250-225)^2}{225} + \frac{(50-75)^2}{75} + \frac{(200-225)^2}{225} + \frac{(100-75)^2}{75}$$

$$\chi^2 = \frac{625}{225} + \frac{625}{75} + \frac{625}{225} + \frac{625}{75} = 2.778 + 8.333 + 2.778 + 8.333 = \mathbf{22.22}$$

3. Hypothesis Decision:

- Degrees of Freedom: $df = (r-1)(c-1) = (2-1)(2-1) = \mathbf{1}$.
- Critical value from χ^2 distribution table at $\alpha = 0.05, df = 1$ is $\mathbf{3.841}$.
- $\mathbf{22.22 \gg 3.841} \implies \mathbf{\text{Reject Null Hypothesis! Gender and News Medium are strongly correlated!}}$

🏛 Step-by-Step Solved Problem: Min-Max Normalization & Z-Score Scaling

Suppose an attribute **Income** ranges from $\min = \$12,000$ to $\max = \$98,000$ with mean $\bar{x} = \$54,000$ and standard deviation $\sigma = \$16,000$. Transform value $v = \$73,600$:

1. Min-Max Normalization to Range $[0.0, 1.0]$:

$$v' = \frac{73600 - 12000}{98000 - 12000} = \frac{61600}{86000} = \mathbf{0.7163}$$

2. Min-Max Normalization to Range $[-1.0, 1.0]$:

$$v' = \left(\frac{61600}{86000}\right)(1 - (-1)) + (-1) = 0.7163(2) - 1 = 1.4326 - 1 = +0.4326$$

3. Z-Score Standardization:

$$z = \frac{v - \bar{x}}{\sigma} = \frac{73600 - 54000}{16000} = \frac{19600}{16000} = +1.225$$

4. Decimal Scaling Normalization:

Since $\max(|v|) = 98,000$, choose $j = 5$ (so that $\frac{98000}{10^5} = 0.98 < 1$):

$$v' = \frac{73600}{10^5} = \mathbf{0.736}$$

Topic 19 & 20: Dimensionality Reduction (PCA) & Solved Exam Questions

🏛 The 5-Step Principal Component Analysis (PCA) Algorithm

1. **Mean-Center Data:** Subtract column mean from each feature vector: $\mathbf{X}_{\text{centered}} = \mathbf{X} - \bar{\mathbf{X}}$.
2. **Compute Covariance Matrix:** $\boldsymbol{\Sigma} = \frac{1}{N-1}\mathbf{X}^T_{\text{centered}}\mathbf{X}_{\text{centered}} \in \mathbb{R}^{p\times p}$.
3. **Compute Eigenvalues & Eigenvectors:** Solve characteristic polynomial $\det(\boldsymbol{\Sigma} - \lambda\mathbf{I}) = 0$ to obtain eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$ and orthogonal eigenvectors $\mathbf{e}_1, \ldots, \mathbf{e}_p$.
4. **Select Top k Principal Components:** Choose $k \ll p$ eigenvectors corresponding to the largest eigenvalues such that Explained Variance Ratio $\geq 95\%$:

$$\text{EVR} = \frac{\sum_{i=1}^{k}\lambda_i}{\sum_{j=1}^{p}\lambda_j} \geq \mathbf{0.95}$$

5. **Project Data:** Transform original p-dimensional data into reduced k-dimensional space: $\mathbf{Y} = \mathbf{X}_{\text{centered}}\mathbf{W}_k$ (where $\mathbf{W}_k = [\mathbf{e}_1, \ldots, \mathbf{e}_k]$).

Q1. Explain Discrete Wavelet Transform (DWT) and how it differs from Discrete Fourier Transform (DFT). (8 Marks)

• **Fourier Transform:** Decomposes a signal into global sine and cosine waves. It has perfect frequency resolution but *zero time localization* (a transient spike in a time series corrupts the entire spectrum).
• **Wavelet Transform:** Uses localized basis functions (wavelets) that possess finite duration in both time and frequency. DWT applies a pyramid of low-pass (approximation/coarse) and high-pass (detail/fluctuation) filters, allowing effective data compression and noise removal without losing transient events.

Q2. Differentiate between Stepwise Forward Selection and Stepwise Backward Elimination in Attribute Subset Selection. (8 Marks)

• **Forward Selection:** Begins with an empty feature set $\emptyset$. At each step, it trains models with each remaining candidate feature and adds the single attribute that maximizes model performance (e.g. lowest AIC/BIC or highest F-statistic) until additions no longer yield significant statistical improvement.
• **Backward Elimination:** Begins with the complete set of all p features. At each step, it removes the single attribute that contributes least to model accuracy (highest p-value) until all remaining features are statistically significant ($p < 0.05$).

Topic 20.1: Advanced Dimensionality Reduction (LDA, SVD & t-SNE)

Reduction Technique	Core Mathematical Objective	Supervision & Linearity
Principal Component Analysis (PCA)	Maximizes projected variance: $\max_{\mathbf{w}} \mathbf{w}^T\boldsymbol{\Sigma}\mathbf{w}$ s.t. $\|\mathbf{w}\| = 1$. Unsupervised linear projection.	Unsupervised • Linear
Linear Discriminant Analysis (LDA)	Maximizes between-class scatter over within-class scatter (Fisher's Criterion): $\max_{\mathbf{w}} \frac{\mathbf{w}^T\mathbf{S}_B\mathbf{w}}{\mathbf{w}^T\mathbf{S}_W\mathbf{w}}$.	**Supervised** • Linear
Singular Value Decomposition (SVD)	Decomposes data matrix $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$. Truncated SVD provides optimal rank-k low-rank approximation (Eckart-Young-Mirsky Theorem).	Unsupervised • Linear
t-SNE (t-Distributed Stochastic Neighbor Embedding)	Minimizes KL-divergence between high-dimensional Gaussian joint probabilities p_{ij} and low-dimensional Student-t probabilities q_{ij}: $\mathcal{L} = \sum p_{ij}\log(p_{ij}/q_{ij})$.	Unsupervised • **Non-linear Manifold**

🏛 Step-by-Step Solved Problem: SVD Matrix Decomposition Calculation

Given rank-1 data matrix $\mathbf{X} = \begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix}$:

1. Compute $\mathbf{X}^T\mathbf{X} = \begin{bmatrix} 3 & 4 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 25 & 25 \\ 25 & 25 \end{bmatrix}$.
2. Eigenvalues of $\mathbf{X}^T\mathbf{X}$: $\det\begin{pmatrix} 25-\lambda & 25 \\ 25 & 25-\lambda \end{pmatrix} = (25-\lambda)^2 - 625 = \lambda^2 - 50\lambda = 0 \implies \boldsymbol{\lambda_1 = 50,\ \lambda_2 = 0}$.
3. Singular value $\sigma_1 = \sqrt{50} = \mathbf{5\sqrt{2}} \approx \mathbf{7.071}$.
4. Right Singular Vector $\mathbf{v}_1$: $(\mathbf{X}^T\mathbf{X} - 50\mathbf{I})\mathbf{v} = 0 \implies \begin{bmatrix} -25 & 25 \\ 25 & -25 \end{bmatrix}\mathbf{v} = 0 \implies \mathbf{v}_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$.
5. Left Singular Vector $\mathbf{u}_1 = \frac{1}{\sigma_1}\mathbf{X}\mathbf{v}_1 = \frac{1}{5\sqrt{2}}\begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix}\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{10}\begin{bmatrix} 6 \\ 8 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$.

$$\mathbf{X} = \mathbf{u_1}\sigma_1\mathbf{v_1^T} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}(\mathbf{5\sqrt{2}})\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 4 & 4 \end{bmatrix} \quad \text{(Exact Rank-1 SVD!)}$$

Q3. Explain Kernel PCA (KPCA) and the Mercer Kernel Trick. (8 Marks)

Standard PCA fails when data points reside on non-linear manifolds (e.g., concentric circles or Swiss roll).
Kernel PCA (Schölkopf 1998): Maps input data into a high-dimensional reproducing kernel Hilbert space $\Phi(\mathbf{x}) \in \mathcal{H}$ and performs linear PCA in that feature space. By **Mercer's Theorem**, the inner products in $\mathcal{H}$ can be computed directly in input space using kernel functions ($K(\mathbf{x}_i, \mathbf{x}_j) = \langle \Phi(\mathbf{x}_i), \Phi(\mathbf{x}_j) \rangle$), such as the Gaussian RBF kernel $K(\mathbf{x}, \mathbf{y}) = \exp(-\gamma \|\mathbf{x} - \mathbf{y}\|^2)$, discovering complex non-linear cluster structures!

Topic 20.2: Advanced Preprocessing & Feature Selection Topologies

🏛 Step-by-Step Solved Problem: Discrete Wavelet Transform (Haar Wavelet) 1D Signal Decomposition

Consider 1D discrete time series signal $\mathbf{X} = [2, 4, 6, 8, 10, 12, 14, 16]$ ($N = 8$):

1. **Level 1 Decomposition:**
   - Averages (Low-pass coarse trend): $\left[\frac{2+4}{2}, \frac{6+8}{2}, \frac{10+12}{2}, \frac{14+16}{2}\right] = [\mathbf{3}, \mathbf{7}, \mathbf{11}, \mathbf{15}]$.
   - Differences (High-pass wavelet details): $\left[\frac{2-4}{2}, \frac{6-8}{2}, \frac{10-12}{2}, \frac{14-16}{2}\right] = [\mathbf{-1}, \mathbf{-1}, \mathbf{-1}, \mathbf{-1}]$.
2. **Level 2 Decomposition (on Averages $[3, 7, 11, 15]$):**
   - Level 2 Averages: $\left[\frac{3+7}{2}, \frac{11+15}{2}\right] = [\mathbf{5}, \mathbf{13}]$.
   - Level 2 Details: $\left[\frac{3-7}{2}, \frac{11-15}{2}\right] = [\mathbf{-2}, \mathbf{-2}]$.
3. **Level 3 Decomposition (on $[5, 13]$):**
   - Level 3 Average: $\frac{5+13}{2} = \mathbf{9}$.
   - Level 3 Detail: $\frac{5-13}{2} = \mathbf{-4}$.

$$\text{Full Haar Wavelet Representation: } [\mathbf{9},\ \mathbf{-4},\ \mathbf{-2}, \mathbf{-2},\ \mathbf{-1}, \mathbf{-1}, \mathbf{-1}, \mathbf{-1}]$$

Data Compression: The detail coefficients are highly sparse, allowing extreme compression by thresholding small details to zero!

Q4. Explain ReliefF Algorithm for Feature Weighting and Selection. (8 Marks)

ReliefF (Kononenko 1994) evaluates the quality of individual attributes according to how well their values distinguish between instances that are near to each other:
1. Randomly sample an instance R_i.
2. Find its k nearest instances belonging to the *same class* (Near Hits H) and k nearest instances belonging to *different classes* (Near Misses M).
3. Update weight $W[A]$ for attribute A: Increase weight if A separates R_i from Near Misses; decrease weight if A separates R_i from Near Hits:

$$\mathbf{W[A] \leftarrow W[A] - \sum_{j=1}^{k} \frac{\text{diff}(A, R_i, H_j)}{m \cdot k} + \sum_{C \neq \text{class}(R_i)} \frac{P(C)}{1 - P(\text{class}(R_i))} \sum_{j=1}^{k} \frac{\text{diff}(A, R_i, M_j(C))}{m \cdot k}}$$

ReliefF successfully captures non-linear feature interactions and scales linearly $O(m \cdot k \cdot p)$!

Topic 20.3: Master University Exam Problem Bank (Part II)

🏛 Solved Problem 2: Pearson Correlation Coefficient with Standardized Variance

Calculate Pearson's correlation coefficient $r_{A,B}$ for 5 observation pairs: $(1, 2), (2, 4), (3, 5), (4, 7), (5, 8)$:

- $\bar{A} = \frac{1+2+3+4+5}{5} = \mathbf{3.0} \qquad \bar{B} = \frac{2+4+5+7+8}{5} = \frac{26}{5} = \mathbf{5.2}$
- Deviations $a_i - \bar{A}$: $[-2, -1, 0, +1, +2] \implies \sum(a_i - \bar{A})^2 = 4 + 1 + 0 + 1 + 4 = \mathbf{10.0}$
- Deviations $b_i - \bar{B}$: $[-3.2, -1.2, -0.2, +1.8, +2.8] \implies \sum(b_i - \bar{B})^2 = 10.24 + 1.44 + 0.04 + 3.24 + 7.84 = \mathbf{22.80}$
- Cross-product $\sum(a_i - \bar{A})(b_i - \bar{B}) = (-2)(-3.2) + (-1)(-1.2) + 0 + (1)(1.8) + (2)(2.8) = 6.4 + 1.2 + 0 + 1.8 + 5.6 = \mathbf{15.0}$

$$\mathbf{r_{A,B} = \frac{15.0}{\sqrt{10.0 \times 22.80}} = \frac{15.0}{\sqrt{228.0}} = \frac{15.0}{15.0997} = +0.9934}$$

$$\text{Conclusion: Exceptionally strong linear correlation } (+0.993) \text{ between attributes A and B!}$$

Q5. Explain the Concept of Data Reduction via Numerosity Reduction (Parametric vs Non-parametric). (8 Marks)

• **Parametric Numerosity Reduction:** Fits the data to a mathematical model (Linear/Multiple Regression, Log-Linear Models) and stores only the learned model parameters (intercept β_0, slope weights β, and residual variance σ^2), completely discarding the raw training records.
• **Non-Parametric Numerosity Reduction:** Does NOT assume any functional form. Uses Histograms (equal-width, equal-frequency, V-optimal), Clustering (replacing groups of points with their cluster centroids), Sampling (SRS, stratified sampling), and Data Cube Aggregation.

Q6. Explain the Box-Cox Power Transformation for Skewed Non-Gaussian Data. (8 Marks)

The **Box-Cox Transformation** is a parametric power transformation that stabilizes variance and transforms highly skewed distributions into an approximate Gaussian bell curve:

$$\mathbf{y}^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \ln(y) & \text{if } \lambda = 0 \end{cases}$$

The parameter λ is estimated via Maximum Likelihood Estimation (MLE). If $\lambda = 1$, no transform is needed; if $\lambda = 0$, a natural log transform is applied; if $\lambda = 0.5$, a square root transform is used.

Topic 20.4: Master University Exam Problem Bank (Part III)

🏛 **Solved Problem 3: Complete 2D Principal Component Analysis (PCA) Eigenvalue Derivation**

Consider 4 centered 2D data points: $\mathbf{x}_1 = (-1, -1)^T, \mathbf{x}_2 = (-1, 1)^T, \mathbf{x}_3 = (1, -1)^T, \mathbf{x}_4 = (1, 1)^T$ with correlation covariance $\mathbf{\Sigma} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$:

1. Characteristic Polynomial:

$$\det(\mathbf{\Sigma} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - \mathbf{3})(\lambda - \mathbf{1}) = \mathbf{0} \implies \lambda_\mathbf{1} = \mathbf{3}, \quad \lambda_\mathbf{2} = \mathbf{1}$$

2. First Principal Component Eigenvector ($\lambda_1 = 3$):

$$(\mathbf{\Sigma} - 3\mathbf{I})\mathbf{e}_1 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies -e_{11} + e_{12} = 0 \implies e_{11} = e_{12}$$

$$\text{Normalized First Principal Component: } \mathbf{e_1} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \approx \begin{bmatrix} 0.7071 \\ 0.7071 \end{bmatrix}$$

3. Explained Variance Ratio:

$$\text{EVR}_\mathbf{1} = \frac{\lambda_\mathbf{1}}{\lambda_\mathbf{1} + \lambda_\mathbf{2}} = \frac{\mathbf{3}}{\mathbf{3} + \mathbf{1}} = \frac{\mathbf{3}}{\mathbf{4}} = \mathbf{75\%} \text{ of total dataset variance captured in 1D!}$$

Q7. Explain the Difference between L1-Norm (Lasso) and L2-Norm (Ridge) Regularization in Regression. (8 Marks)

• **Ridge Regression (L_2):** Penalizes squared magnitude of coefficients ($\lambda \sum \beta_j^2$). Shrinks weights smoothly toward zero, preventing multicollinearity, but *never sets weights strictly to zero* (does not perform feature selection).
• **Lasso Regression (L_1):** Penalizes absolute value of coefficients ($\lambda \sum |\beta_j|$). Due to the diamond geometric contour of the L_1 ball, optimal solutions hit coordinate axes, driving irrelevant feature weights **strictly to zero** ($\beta_j = 0$), performing automatic sparse feature selection!

Topic 20.5: Master University Exam Problem Bank (Part IV)

🏛️ Solved Problem 4: Complete Chi-Square Contingency Test with 3×3 Degrees of Freedom

Test the relationship between Customer Age Group $(\text{Youth, Middle, Senior})$ and Purchase Frequency (Low, Med, High):

Age \ Frequency	Low	Medium	High	Total
Youth	$O_{11} = 50$	$O_{12} = 30$	$O_{13} = 20$	100
Middle	$O_{21} = 30$	$O_{22} = 50$	$O_{23} = 20$	100
Senior	$O_{31} = 20$	$O_{32} = 20$	$O_{33} = 60$	100
Total	100	100	100	$\mathbf{N = 300}$

1. Expected Frequencies: $E_{ij} = \frac{100 \times 100}{300} = \mathbf{33.33}$ for all cells.

2. Compute χ^2 Statistic:

$$\chi^2 = \frac{(50-33.33)^2 + (30-33.33)^2 + (20-33.33)^2 + (30-33.33)^2 + (50-33.33)^2 + (20-33.33)^2 + (20-33.33)^2 + (20-33.33)^2 + (60-33.33)^2}{33.33}$$

$$\chi^2 = \frac{277.89 + 11.09 + 177.69 + 11.09 + 277.89 + 177.69 + 177.69 + 177.69 + 711.29}{33.33} = \frac{1999.98}{33.33} \approx \mathbf{60.00}$$

3. Decision: $df = (3-1)(3-1) = 4$. Critical $\chi^2_{0.05,4} = 9.488$. $\mathbf{60.00} \gg \mathbf{9.488} \implies \text{Reject Independence!}$

Q8. Explain Principal Component Regression (PCR) and Partial Least Squares (PLS). (8 Marks)

• **PCR:** A two-step pipeline that first performs unsupervised PCA on the predictors $\mathbf{X}$ to extract the top k principal components $\mathbf{Z}$, and then fits an ordinary least squares regression model predicting $\mathbf{y}$ from $\mathbf{Z}$. Prevents multicollinearity, but because PCA is unsupervised, the chosen components may not necessarily be the most predictive of $\mathbf{y}$!
• **PLS (Partial Least Squares): Supervised Dimensionality Reduction** that finds latent orthogonal components that maximize the covariance between $\mathbf{X}$ and target $\mathbf{y}$, ensuring retained components have maximal predictive power!

Q9. Detail the Discretization of Numeric Attributes via Maximum Entropy Splitting. (8 Marks)

Maximum Entropy Discretization partitions a continuous attribute into k intervals such that the Shannon entropy over all bins is maximized. This creates bins that contain approximately equal numbers of data points (equi-frequency), preventing low-density outlier intervals from distorting decision tree splits and probability estimates in Naive Bayes.

Topic 20.6: Complete Solved Laboratory Problem Bank on Data Cleaning

🏛️ Solved Numerical 5: Multi-Attribute Missing Imputation via k-Nearest Neighbors (k-NN)

Consider 5 customer profiles with attributes Age and Income (in $\backslash k). A new record$X = (30, ?)$ has missing Income:

- $P_1 = (25, \$40\text{k}),\ P_2 = (32, \$60\text{k}),\ P_3 = (28, \$55\text{k}),\ P_4 = (45, \$90\text{k}),\ P_5 = (31, \$58\text{k}).$

1. Calculate Euclidean Distances on Normalized Age $(\min = 25, \max = 45)$:

- $d(X, P_1) = \frac{|30-25|}{20} = \frac{5}{20} = 0.25$
- $d(X, P_2) = \frac{|30-32|}{20} = \frac{2}{20} = 0.10$
- $d(X, P_3) = \frac{|30-28|}{20} = \frac{2}{20} = 0.10$
- $d(X, P_4) = \frac{|30-45|}{20} = \frac{15}{20} = 0.75$
- $d(X, P_5) = \frac{|30-31|}{20} = \frac{1}{20} = \mathbf{0.05}\ (\textbf{Nearest!})$

2. Imputation with $k = 3$ Neighbors (P_5, P_2, P_3):

$$\text{Imputed Income} = \frac{\mathbf{\$58}\text{k} + \mathbf{\$60}\text{k} + \mathbf{\$55}\text{k}}{\mathbf{3}} = \frac{\mathbf{\$173}\text{k}}{\mathbf{3}} = \mathbf{\$57.67}\text{k}$$

🏛 Solved Problem 6: Decimal Scaling & Robust Outlier MAD Normalization

For dataset $X = \{10, 12, 14, 15, 16, 18, 100\}$ with $\text{Median} = 15$:

- Deviations from median $|x_i - 15|$: $\{5, 3, 1, 0, 1, 3, 85\}$.
- Sorted deviations: $\{0, 1, 1, 3, 3, 5, 85\} \implies \text{MAD} = \mathbf{3.0}$.
- Robust Z-score for outlier 100: $M_i = \frac{0.6745(100-15)}{3.0} = \frac{0.6745(85)}{3.0} = \frac{57.33}{3.0} = \mathbf{19.11} \gg \mathbf{3.5} \implies \text{Extreme Outlier!}$

🏛 Solved Problem 7: Data Discretization by ChiMerge (χ^2-based Interval Merging)

ChiMerge (Kerber 1992) is a bottom-up supervised discretization algorithm:

1. Sort continuous attribute values and place each distinct value into its own interval.
2. For each pair of adjacent intervals, compute the χ^2 statistic with respect to class labels.
3. Identify the pair of adjacent intervals with the **smallest χ^2 value** (indicating that their class distributions are most similar).
4. Merge that pair into a single interval.
5. Repeat steps 2-4 until all adjacent interval pairs have $\chi^2 \geq \text{threshold}$ (or until target number of intervals is reached).

🏛 Solved Problem 8: Feature Extraction using Linear Discriminant Analysis (LDA) Projection

For 2 classes with means $\mu_1 = (1, 2)^T, \mu_2 = (3, 4)^T$ and shared covariance $\mathbf{\Sigma} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$:

$$\mathbf{w} \propto \mathbf{\Sigma}^{-1}(\mu_1 - \mu_2) = \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -2 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\text{Optimal 1D Fisher Projection Line: } \mathbf{y} = -\mathbf{x_1} - \mathbf{x_2}$$

🏛 Solved Numerical 9: Non-Negative Matrix Factorization (NMF) for Topic Extraction

Unlike PCA (which allows negative weights), **NMF** factorizes data matrix $\mathbf{V} \approx \mathbf{WH}$ where all matrices are non-negative ($V_{ij} \geq 0, W_{ik} \geq 0, H_{kj} \geq 0$). Solves Frobenius norm minimization:

$$\min_{\mathbf{W},\mathbf{H}} ||\mathbf{V} - \mathbf{WH}||_F^2 \quad \text{subject to } W \geq 0, H \geq 0$$

Multiplicative update rules: $H_{aj} \leftarrow H_{aj}\frac{(W^TV)_{aj}}{(W^TWH)_{aj}}$ and $W_{ia} \leftarrow W_{ia}\frac{(VH^T)_{ia}}{(WHH^T)_{ia}}$. Produces additive, interpretable parts-based representations!

Q10. Explain the Difference between Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). (8 Marks)

• **MCAR:** The probability of an attribute value being missing is completely independent of both observed and unobserved data (e.g., test tube dropped by accident). Dropping tuples causes zero statistical bias.
• **MAR:** The missingness depends systematically on *observed* variables but not on the missing value itself (e.g., males are less likely to report depression scores, but conditional on gender, missingness is random). Multiple imputation (MICE) is valid.
• **MNAR (Non-ignorable):** The missingness depends directly on the unobserved value itself (e.g., high-income executives refuse to disclose income on surveys). Requires explicit modeling of the missingness mechanism!

🏛 Solved Problem 11: Z-Score Standardization vs Min-Max Normalization Trade-Off Analysis

Consider a dataset with severe outliers: $X = \{10, 11, 12, 13, 14, 15, 1000\}$:

- **Min-Max Scaling to** $[0, 1]$: $x_1 = \frac{10-10}{1000-10} = 0.000$; $x_5 = \frac{14-10}{990} = 0.004$; $x_7 = \frac{1000-10}{990} = 1.000$. The normal cluster $\{10..15\}$ is crushed into a tiny range $[0.000, 0.005]$, completely destroying feature variance!
- **Z-Score Standardization:** Outlier 1000 pulls the mean to $\bar{x} = 154.1$ and $\sigma = 373.1$.
- **Robust Standardization (Median & IQR):** $x' = \frac{x-13}{3.0}$. Points $\{10..15\}$ span $[-1.0, +0.67]$, preserving high variance, while outlier 1000 becomes $+329.0$ without compressing the cluster!

Q12. Explain Data Cleaning for Inconsistent Codes and Duplicate Elimination (Record Linkage). (8 Marks)

Real-world databases contain duplicate entities recorded with differing syntax (e.g. `IBM Corp`, `International Business Machines`, `I.B.M.`):
1. **String Similarity Metrics:** Levenshtein Edit Distance, Jaro-Winkler Metric ($d_J = \frac{1}{3}\left(\frac{m}{|s_1|} + \frac{m}{|s_2|} + \frac{m-t}{m}\right)$), q-gram token overlapping.
2. **Blocking (Canopy Clustering):** Divides N records into overlapping canopies using fast phonetic hash keys (Soundex / Metaphone) to avoid $O(N^2)$ pairwise comparisons.
3. **Merge/Purge Pipeline:** Merges duplicate records into a golden master record with unified primary keys.

🏛 **Solved Problem 13: Principal Component Analysis (PCA) Variance Proof & SVD Relationship**

Prove that the principal component directions are given by the eigenvectors of the sample covariance matrix $\mathbf{\Sigma}$:

- Objective: Maximize projected variance $\mathbf{w}^T\mathbf{\Sigma}\mathbf{w}$ subject to $\|\mathbf{w}\|_2^2 = \mathbf{w}^T\mathbf{w} = 1$.
- Lagrangian: $\mathcal{L}(\mathbf{w}, \lambda) = \mathbf{w}^T\mathbf{\Sigma}\mathbf{w} - \lambda(\mathbf{w}^T\mathbf{w} - 1)$.
- Taking gradient with respect to $\mathbf{w}$ and setting to zero:

$$\nabla_{\mathbf{w}}\mathcal{L} = 2\mathbf{\Sigma}\mathbf{w} - 2\lambda\mathbf{w} = 0 \implies \mathbf{\Sigma}\mathbf{w} = \lambda\mathbf{w}$$

- This is the standard eigenvector equation! Multiplying by $\mathbf{w}^T$: $\mathbf{w}^T\mathbf{\Sigma}\mathbf{w} = \lambda\mathbf{w}^T\mathbf{w} = \lambda$.
- $$\text{Maximum variance is achieved by choosing the eigenvector corresponding to the largest eigenvalue } \lambda_{\max}!$$

Q13. Explain the V-Optimal Histogram Construction for Numerosity Reduction. (8 Marks)

A **V-Optimal Histogram** partitions N data values into B buckets such that the weighted variance of the bucket estimates is minimized across all possible partitionings:

$$\mathbf{\min \sum_{j=1}^{B} \sum_{i \in Bucket_j} (x_i - \bar{x}_j)^2}$$

Dynamic programming computes the globally optimal bucket boundaries in $O(B \cdot N^2)$ time. V-Optimal histograms provide the most statistically accurate selectivity estimation for database query optimizers!

🏛 **Solved Problem 14: Kernel Density Estimation (KDE) for Continuous Density Smoothing**

For N observations, **Kernel Density Estimation** computes continuous probability density function $\hat{f}(x)$ with Gaussian kernel $K(u) = \frac{1}{\sqrt{2\pi}}e^{-u^2/2}$ and bandwidth h:

$$\mathbf{\hat{f}(x) = \frac{1}{Nh}\sum_{i=1}^{N} K\left(\frac{x - x_i}{h}\right) = \frac{1}{Nh\sqrt{2\pi}}\sum_{i=1}^{N} \exp\left(-\frac{(x - x_i)^2}{2h^2}\right)}$$

Bandwidth Selection (Silverman's Rule of Thumb): $h = 0.9\min\left(\hat{\sigma}, \frac{\text{IQR}}{1.34}\right)N^{-1/5}$. Prevents oversmoothing (high bias) and undersmoothing (spurious variance)!

🏛 **Solved Problem 15: Pearson Correlation vs Covariance Proof & Standardized Derivation**

Prove that Pearson's correlation coefficient $r_{X,Y} \in [-1, +1]$ using Cauchy-Schwarz Inequality:

$$\mathbf{|\mathrm{Cov}(X, Y)|^2 = |\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]|^2 \le \mathbb{E}[(X - \mu_X)^2] \cdot \mathbb{E}[(Y - \mu_Y)^2] = \sigma_X^2\sigma_Y^2}$$
$$\mathbf{\frac{|\mathrm{Cov}(X, Y)|}{\sigma_X\sigma_Y} \le 1 \implies -1 \le r_{X,Y} \le +1}$$

Q14. Explain Missing Value Handling in Time Series Streams (Forward Fill, Linear Interpolation, Splines). (8 Marks)

• **Forward Fill (Last Observation Carried Forward - LOCF):** Replaces missing timestamp t with the most recent valid observation $t-1$. Appropriate for static sensor readings.
• **Linear / Cubic Spline Interpolation:** Fits a piecewise polynomial curve through surrounding valid points, preserving continuous trajectory curvature.

🏛 **Solved Problem 16: Discrete Fourier Transform (DFT) vs Wavelet Transform Signal Compression**

For a 1D sequence of length $N = 2^n$, DFT projects data onto complex sinusoids $X_k = \sum_{n=0}^{N-1} x_n e^{-i2\pi kn/N}$ ($O(N\log N)$ FFT). In contrast, Wavelet transform uses compact dyadic wavelets $\psi_{j,k}(t) = 2^{j/2}\psi(2^j t - k)$, achieving linear $O(N)$ computational complexity and retaining precise time-frequency localization of transient data spikes!

Q15. Explain Stratified Sampling vs Simple Random Sampling (SRS) in Imbalanced Data Mining. (8 Marks)

• **Simple Random Sampling (SRS):** Every record has equal probability $1/N$ of selection. On highly imbalanced datasets (e.g. 99% legitimate, 1% fraud), SRS risks drawing zero fraud instances!
• **Stratified Sampling:** Divides the dataset into non-overlapping strata (classes $C_1, \ldots, C_k$) and draws samples from each stratum proportionally (or equally via oversampling), guaranteeing that rare minority classes are preserved in training subsets!

🏛 Solved Problem 17: Multi-Class Linear Discriminant Analysis (LDA) Between-Class Scatter Matrix Derivation

For C classes with means μ_c and global mean μ:

$$\mathbf{S}_B = \sum_{c=1}^{C} N_c(\mu_c - \mu)(\mu_c - \mu)^T \qquad \mathbf{S}_W = \sum_{c=1}^{C}\sum_{i\in c}(\mathbf{x}_i - \mu_c)(\mathbf{x}_i - \mu_c)^T$$

Maximizing Rayleigh quotient $\frac{\mathbf{w}^T\mathbf{S}_B\mathbf{w}}{\mathbf{w}^T\mathbf{S}_W\mathbf{w}}$ yields generalized eigenvalue problem $\mathbf{S}_W^{-1}\mathbf{S}_B\mathbf{w} = \lambda\mathbf{w}$. Number of non-zero discriminant directions is at most $\min(\mathbf{p}, \mathbf{C}-\mathbf{1})$!

🏛 Solved Problem 18: Quantile Normalization for High-Throughput Genomics & Microarrays

Quantile Normalization aligns empirical distributions across multiple experimental batches:

1. Sort values in each column independently.
2. Compute the mean across rows for the sorted values.
3. Replace each sorted value with the row average.
4. Restore columns back to their original rank orders, ensuring all columns have identical empirical distributions!

🏛 Solved Problem 19: Discrete Cosine Transform (DCT) vs PCA Data Compression

The **Discrete Cosine Transform (DCT)** expresses a finite sequence of data points in terms of a sum of cosine functions oscillating at different frequencies:

$$\mathbf{X_k} = \sum_{\mathbf{n=0}}^{\mathbf{N-1}} \mathbf{x_n} \cos\left[\frac{\pi}{\mathbf{N}}\left(\mathbf{n}+\frac{\mathbf{1}}{\mathbf{2}}\right)\mathbf{k}\right]$$

• **Energy Compaction Property:** DCT concentrates most of the signal energy into the first few low-frequency transform coefficients.
• **PCA vs DCT:** PCA requires computing data covariance matrices $O(Np^2)$ and solving eigenvalue polynomials. DCT uses fixed data-independent cosine basis functions, allowing $O(N\log N)$ fast FFT computation (used in JPEG image compression and MP3 audio encoding)!

🏛 Solved Problem 20: Covariance & Correlation Matrix Derivation for 3-Variable Dataset

Given 3 attributes A, B, C with mean-centered vectors $\mathbf{a} = (-1, 0, 1)^T, \mathbf{b} = (2, -1, -1)^T, \mathbf{c} = (0, 1, -1)^T$:

- Variances: $s_A^2 = \frac{1+0+1}{2} = 1.0;\ s_B^2 = \frac{4+1+1}{2} = 3.0;\ s_C^2 = \frac{0+1+1}{2} = 1.0.$
- Covariances:
  - $\text{Cov}(A, B) = \frac{(-1)(2)+(0)(-1)+(1)(-1)}{2} = \frac{-2+0-1}{2} = -\mathbf{1.5}$
  - $\text{Cov}(A, C) = \frac{(-1)(0)+(0)(1)+(1)(-1)}{2} = \frac{-1}{2} = -\mathbf{0.5}$
  - $\text{Cov}(B, C) = \frac{(2)(0)+(-1)(1)+(-1)(-1)}{2} = \frac{-1+1}{2} = \mathbf{0.0}$

$$\mathbf{\mathbf{\Sigma} = \begin{bmatrix} 1.0 & -1.5 & -0.5 \\ -1.5 & 3.0 & 0.0 \\ -0.5 & 0.0 & 1.0 \end{bmatrix} \qquad \mathbf{R} = \begin{bmatrix} 1.0 & -0.866 & -0.500 \\ -0.866 & 1.0 & 0.0 \\ -0.500 & 0.0 & 1.0 \end{bmatrix}}$$

🏛 Solved Problem 21: Multi-Dimensional Data Transformation & Wavelet Thresholding

In Wavelet soft thresholding: $\hat{w} = \text{sign}(w)(|w| - \lambda)_+$ with Universal threshold $\lambda = \sigma\sqrt{2\ln N}$. Discards high-frequency stochastic noise coefficients while preserving sharp edge boundaries in spatial signals!

🏛 Solved Problem 22: Singular Value Decomposition (SVD) Matrix Approximations

For any rectangular matrix $\mathbf{X} \in \mathbb{R}^{n\times d}$, SVD decomposes $\mathbf{X} = \mathbf{U\Sigma V}^T$ where $\mathbf{U} \in \mathbb{R}^{n\times n}$ and $\mathbf{V} \in \mathbb{R}^{d\times d}$ are orthogonal matrices and $\mathbf{\Sigma} \in \mathbb{R}^{n\times d}$ contains sorted singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

- **Eckart-Young-Mirsky Theorem:** The rank-k truncated matrix $\mathbf{X}_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ is the provably optimal rank-k approximation minimizing both Frobenius and spectral matrix norms!
- Used in Latent Semantic Indexing (LSI), Netflix collaborative filtering, and facial image compression!

🏛️ **Solved Problem 23: Fast Fourier Transform (FFT) vs Wavelet Energy Compaction Proof**

For discrete signal $x = [1, 3, 5, 7]$:

- Haar Wavelet level 1 averages: $a_1 = [\frac{1+3}{\sqrt{2}}, \frac{5+7}{\sqrt{2}}] = [2\sqrt{2}, 6\sqrt{2}] \approx [2.828, 8.485]$.
- Haar Wavelet level 1 details: $d_1 = [\frac{1-3}{\sqrt{2}}, \frac{5-7}{\sqrt{2}}] = [-\sqrt{2}, -\sqrt{2}] \approx [-1.414, -1.414]$.
- Haar level 2 average: $a_2 = \frac{2\sqrt{2}+6\sqrt{2}}{\sqrt{2}} = \mathbf{8.0}$.
- Haar level 2 detail: $d_2 = \frac{2\sqrt{2}-6\sqrt{2}}{\sqrt{2}} = -\mathbf{4.0}$.
- Transformed Wavelet vector: $\mathbf{W} = [\mathbf{8.0}, -\mathbf{4.0}, -\mathbf{1.414}, -\mathbf{1.414}]$. Energy is perfectly conserved: $\|W\|^2 = 64 + 16 + 2 + 2 = 84 = \|x\|^2 = 1 + 9 + 25 + 49 = 84$!